

Data Model Documentation

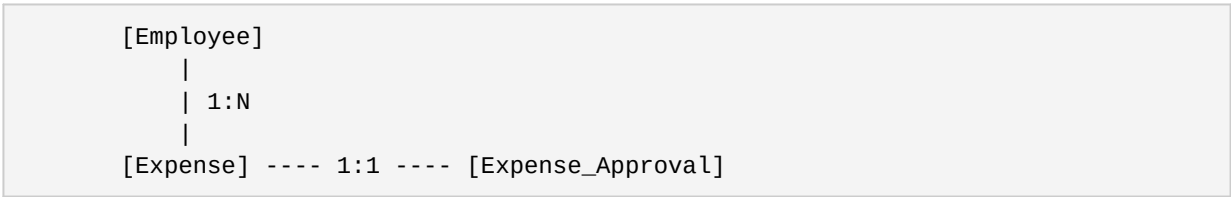
Overview

The expense management application manages the core domains of employees and expenses. The application allows employees to submit expenses for reimbursement and administrators to approve or reject these expenses. This system ensures that all expenses are tracked, approved, and reimbursed efficiently.

The system manages the following core domains:

- **Employee**: Responsible for submitting expenses, with a one-to-many relationship with expenses.
- **Expense**: Represents the expenses submitted by employees, with a one-to-one relationship with approval records.

Entity Relationship Diagram



Entities

Employee

Purpose: Represents an employee who can submit expenses for reimbursement.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the employee
name	VARCHAR	Yes	Max 255 chars	Employee's name
email	VARCHAR	Yes	Unique	Employee's email address
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Field Name	Type	Required	Constraints	Description
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many Expense (1:N): Inverse via Expense.employee_id

Validation Rules:

- email must be non-empty and unique across all records

Expense

Purpose: Represents an expense submitted by an employee for reimbursement.

Table/Collection Name: Expense

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the expense
employee_id	UUID	Yes	Foreign key	Employee who submitted the expense
amount	DECIMAL	Yes	Min 0	Expense amount
description	VARCHAR	No	Max 255 chars	Expense description
submitted_at	TIMESTAMP	Yes	Auto-set on create	Expense submission time
status	VARCHAR	Yes	Default 'PENDING'	Expense status

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references Employee.id
- Has one Expense_Approval (1:1): Inverse via Expense_Approval.expense_id

Validation Rules:

- employee_id must reference an existing employee
- amount must be a positive number
- description must be at most 255 characters if provided

Expense_Approval

Purpose: Represents the approval status of an expense.

Table/Collection Name: Expense_Approval

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier for the approval record
expense_id	UUID	Yes	Foreign key	Expense being approved
approver_id	UUID	Yes	Foreign key	Employee who approved the expense
approved_at	TIMESTAMP	No		Approval timestamp
approval_status	VARCHAR	Yes	Default 'PENDING'	Approval status

Relationships:

- Belongs to Expense (N:1): Foreign key expense_id references Expense.id
- Belongs to Employee (N:1): Foreign key approver_id references Employee.id

Validation Rules:

- expense_id must reference an existing expense
- approver_id must reference an existing employee
- approval_status must be either 'APPROVED' or 'REJECTED'

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
Employee	id	Expense	employee_id	1:N	CASCADE
Expense	id	Expense_Approval	expense_id	1:1	CASCADE
Employee	id	Expense_Approval	approver_id	N:1	CASCADE

Unique Constraints

- `Employee(email)`: Ensures no duplicate email addresses

Data Validation Rules

Business Rules:

- **Employee email uniqueness:** The `email` field in the `Employee` entity must be unique across all records.
- **Positive expense amount:** The `amount` field in the `Expense` entity must be a positive number.
- **Description length limit:** The `description` field in the `Expense` entity must be at most 255 characters if provided.

Format Rules:

- **Email format:** The `email` field in the `Employee` entity must match the standard email format regex.
- **Date format:** The `submitted_at` and `approved_at` fields in the `Expense` and `ExpenseApproval` entities must be in the YYYY-MM-DD format.

Data Lifecycle

Creation

- A new `Employee` is created when a new user registers.
- Default status for `Expense` is `PENDING`.
- `created_at` and `updated_at` are automatically set to the current timestamp on creation.

Updates

- `Employee.email` can only be updated by the employee themselves or an administrator.
- `updated_at` is automatically updated on any modification to an `Employee` or `Expense`.

Archival / Soft Delete

- Records are not archived but marked with a `deleted_at` timestamp if soft delete is required.

Indexes

- Primary key: `id`
- Foreign keys: `employee_id`, `expense_id`, `approver_id`
- Search fields: `name`, `email` (for common queries)

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "created_at": "2023-10-01T10:00:00Z",
  "updated_at": "2023-10-01T10:00:00Z"
}
```

Expense

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "amount": 100.00,
  "description": "Business lunch",
  "submitted_at": "2023-10-01T12:00:00Z",
  "status": "PENDING"
}
```

Expense Approval

```
{
  "id": "789e1234-5678-9abc-def0-123456789abc",
  "expense_id": "123e4567-e89b-12d3-a456-426614174000",
  "approver_id": "550e8400-e29b-41d4-a716-446655440000",
  "approved_at": "2023-10-02T09:00:00Z",
  "approval_status": "APPROVED"
}
```